A set containing the negation of one of its consequences is inconsistent

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: proof text verbatim including the line that cites ax:lnot2 first; no case analysis needed since the hypothesis is literal membership.

Editorial note. The label prop:explicit-inc names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Negation axiom not-A -> (A -> falsum). Propositional-logic part, axiom ax:lnot2. Line 994: the conditional chain the proof instantiates.
- Modus ponens admissibility. This part, \ollabel{prop:mp} (source line 638). Line 995: 'two applications' close both contractions.

Source reference. Original proposition prop-mp: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 639–639. Prerequisite pointer invoked at proof line(s) 995; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lnot2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 503–503. Prerequisite pointer invoked at proof line(s) 994; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If* $\Gamma \vdash !A$ *and* $\neg !A \in \Gamma$*, then* Γ *is inconsistent.*

Proof. $\Gamma \vdash \neg !A \to (!A \to \bot)$ by ax-lnot2. $\Gamma \vdash \bot$ by two applications of prop-mp. □